A Bit Too Approximate: A Universal Zero-Vector Forgery against Higher-Bit Approximate-ISIS

Martin Feussner
Selmer Center, University of Bergen
martin.feussner@uib.no

14 October 2026

This audit used the OpenAI model Daybreak Blue at the ultra reasoning level. OpenAI Codex generated and selected this paper's title and intentionally chose "A Bit Too Approximate" as wordplay on the Higher-Bit Approximate-ISIS scheme name and on the verifier radius becoming too permissive. OpenAI Codex performed AI-assisted cryptanalysis, implementation, experimentation, and technical writing. A separately tasked AI agent performed hostile adversarial review, and a fresh-context AI agent independently reproduced the result from the pinned paper. The final exact-target attack-history recheck and independent Publication Reviewer gate passed, and the package is classified as `VERIFIED_ATTACK`. No independent human verification was performed or required.

Abstract

We analyze Construction 1 of Le Dévéhat, Shizuya, and Hasegawa's higher-bit Approximate-ISIS signature proposal, IACR ePrint 2022/1144. Its ordinary verifier accepts x when

$$\|x\|_2 \le \beta \quad \text{and} \quad \left\|b^d Ax - H(m)\right\|_2 \le \alpha.$$

Section 6.4 says parameter selection must respect $\|x\|_2 \le s$ and $\|e\|_2 \le b^\ell \sigma + \sqrt{n}\, b^d s$. Hence every threshold completion following that prescription has $\alpha \ge \sqrt{n}\, b^d s$. For all three Figure 3 columns labelled "This work," $b^d s > b^k - 1$, while the paper defines $k = \lceil \log_b q \rceil$ and therefore $q \le b^k$. The required error radius already exceeds $\sqrt{n}(q-1)$, an upper bound on the Euclidean norm of every canonical vector in $\mathbb{Z}_q^n$. At $x = 0$ the public-matrix term vanishes, so the same public zero vector is accepted for every message and every public key, with no signing query or secret information.

Full-dimension paper-level experiments covered all three advertised rows, 64 fresh messages per row, and both canonical and centered residue lifts. All 384 target checks accepted on one worker in 11.40 seconds with 61,072 KiB peak resident memory. Precommitted in-domain diagnostics exercised the matrix, hash, norm, and comparison paths. The paper does not publish concrete α, β, or σ values, a lift, hash serialization, complete Ring mapping, or its cited private implementation. We therefore claim neither an implementation replay nor honest signing. The result is quantified only over Construction 1 threshold completions that obey Section 6.4.

1 Introduction

Approximate trapdoors permit a hash-and-sign construction to return a short preimage whose image is close, rather than equal, to a message-dependent target. Chen, Genise, and Mukherjee introduced the Approx-ISIS formulation and the corresponding approximate-trapdoor signature framework [2]. Le Dévéhat, Shizuya, and Hasegawa later proposed retaining only higher bits of the public matrix to reduce key and signature sizes [1]. Their Construction 1 follows the earlier hash-and-sign pattern while adding a b^d approximation factor.

Approximation must remain smaller than the target space. Chen, Genise, and Mukherjee explicitly identify the generic zero-vector boundary: once the allowed error covers the target, the zero preimage is a trivial solution. We make no novelty claim for that observation. The issue here is its concrete interaction

with the later paper's own parameter prescription. Section 6.4 adds a term $\sqrt{n}\, b^d s$ to the required error bound. At every advertised "This work" row, the scalar $b^d s$ alone exceeds the largest canonical coordinate $q-1$ by a factor between 3.89 and 67.63.

The resulting attack is a universal fresh-message forgery:

- choose any message that has not been submitted to a signing oracle;
- output the length-m all-zero vector; and
- make no signing query and use no secret-key information.

It applies pointwise to every public matrix and every random-oracle output. The experiment is useful for confirming the row transcription, full dimensions, modular arithmetic, both natural lift conventions, and live verification paths; the security conclusion itself is algebraic.

Review status. Primary reproduction, hostile adversarial review, independent fresh-context reproduction, the final exact-target attack-history recheck, and independent Publication Review passed. All required publication gates are complete, and the result is classified as `VERIFIED_ATTACK`.

2 Target and claim boundary

2.1 Pinned target

The target is the PDF of *On the Higher-bit Version of Approximate Inhomogeneous Short Integer Solution Problem*, IACR ePrint 2022/1144, by Anaëlle Le Dévéhat, Hiroki Shizuya, and Shingo Hasegawa [1]. The pinned PDF has SHA-256 digest

`cbbbb33cacef911548d52de7990eff74479dbb2e4ab9467974770107458cabeb.`

Its text extraction has SHA-256 digest

`b5b60db7b57c6a0cee37cd0a49a2a5039d8ed0cb7d7ee9ba3c9a19ae909a5bef.`

The audit uses the mathematical construction and the three columns marked "This work" in Figure 3.

2.2 Ordinary verification relation

Construction 1 models $H : \{0,1\}^* \to \mathbb{Z}_q^n$ as a random oracle. The public key is a matrix

$$A \in \mathbb{Z}_{q/b^d}^{n \times m},$$

and a signature is a vector $x \in \mathbb{Z}_{q/b^d}^m$. The verifier accepts exactly when

$$\|x\|_2 \le \beta, \qquad \left\|b^d A x - H(m)\right\|_2 \le \alpha. \tag{1}$$

The paper uses the Euclidean norm by default. It does not specify which integer representative of the second modular vector is normed. We therefore analyze both common fixed conventions: canonical representatives in $\{0, \ldots, q-1\}$ and centered representatives around zero. Definition 8's existential error vector is no stricter than taking a centered representative.

Section 6.4 derives the signing bounds

$$\|x\|_2 \le s, \qquad \|e\|_2 \le b^\ell \sigma + \sqrt{n}\, b^d s, \tag{2}$$

and immediately states that these bounds must be respected when setting the underlying parameters α and β. We formalize a *Section 6.4-compliant completion* as any choice satisfying

$$\beta \ge s, \qquad \alpha \ge b^\ell \sigma + \sqrt{n}\, b^d s. \tag{3}$$

The claim in this paper is limited to such completions. It does not silently replace α or β with empirical norm entries from Figure 3.

2.3 Unavailable concrete profile

Figure 3 prints τ, described as the Gaussian width of the trapdoor R, rather than the σ appearing in Equation (2). It gives no concrete α or β, threshold-rounding rule, residue lift, hash-to-vector mapping, byte order, key encoding, or signature encoding. The proof-of-concept source is attributed to private communication. The table also relies on a Ring implementation without defining a complete mapping from the matrix construction to that structured representation.

These omissions prevent an exact implementation replay. They do not create an ambiguity in the quantified claim: increasing either radius preserves acceptance, $b^\ell\sigma$ is nonnegative, and the conservative lower term $\sqrt{n}\,b^d s$ already covers the target space.

3 Universal zero-vector forgery

Lemma 1 (Target-space bounds)**.** *For $u \in \mathbb{Z}_q^n$, the Euclidean norm of its canonical representative is at most $\sqrt{n}(q-1)$. The norm of its centered representative is at most $\sqrt{n}\lfloor q/2 \rfloor$.*

Proof. Every canonical coordinate lies between 0 and $q-1$, so the sum of squared coordinates is at most $n(q-1)^2$. Every centered coordinate has absolute value at most $\lfloor q/2 \rfloor$. Taking square roots gives the bounds. □

Theorem 1 (Universal forgery)**.** *Suppose Construction 1 uses a Section 6.4-compliant threshold completion and a parameter row satisfying $b^d s > b^k - 1$, where $k = \lceil \log_b q \rceil$. Then the length-$m$ vector $x = 0$ is accepted for every message and every public matrix under either the canonical or centered lift.*

Proof. The definition of k implies $q \le b^k$. From Equation (3) and $\sigma \ge 0$,

$$\alpha \ge \sqrt{n}\,b^d s > \sqrt{n}(b^k - 1) \ge \sqrt{n}(q-1).$$

At $x = 0$, $\|x\|_2 = 0 \le \beta$ and $b^d Ax = 0$ for every A. The second verification value is therefore the norm of $-H(m)$ in $\mathbb{Z}_q^n$. The lemma places its canonical norm at most at $\sqrt{n}(q-1)$; the centered bound is no larger. Both inequalities in Equation (1) hold for every $H(m)$. □

The attack algorithm is constant: output 0^m. It needs the public parameter length m but need not inspect A, query a signer, or know a secret key. The message may be selected after the public key is generated and can be any fresh attacker-chosen string. This is an efficient existential forgery and, because the same signature works for every message, a universal forgery.

4 Affected advertised rows

Table 1 transcribes the three Figure 3 columns marked "This work." Algorithm 2 states that its optimization requires $q = b^k$, giving $q = 65536$ in each executable row. The proof above needs only the weaker relation $q \le b^k$.

Row	n	b	k	$\ell = d$	m	s	$b^d s$	$b^d s/(b^k-1)$
75.0-bit AISIS	512	2	16	11	3584	1453.0	2,975,744.0	45.4069
155.4-bit AISIS	1024	2	16	11	7168	2163.9	4,431,667.2	67.6229
153.9-bit AISIS	1024	4	8	3	7168	3989.3	255,315.2	3.8959

Table 1: Every advertised "This work" row satisfies the theorem by a wide margin. The claimed AISIS bit costs are labels from Figure 3, not security estimates produced by this audit.

The minimum scalar margin is about 3.8959. Therefore no reasonable rounding of the one-decimal s entries can close the inequality. The additive $b^\ell\sigma$ term, omitted from the conservative calculation, can only increase α.

5 Executable reproduction

5.1 Design

The public reproducer is written solely with the Python standard library. For each affected row it:

1. generates a fresh uniform matrix at the full advertised $n \times m$ dimensions in $\mathbb{Z}_{q/b^d}$;
2. derives multiple distinct fresh messages and maps each to $\mathbb{Z}_q^n$ through a domain-separated SHAKE256 random-oracle adapter;
3. evaluates $b^d Ax$ with a general matrix-vector routine on the length-m zero signature, with no special zero shortcut;
4. invokes the two inequalities of the ordinary verifier under both canonical and centered lifts; and
5. records machine-readable row parameters, matrix and message-set digests, exact norm comparisons, query counts, and diagnostics.

The product for an unchanged matrix and signature is reused across messages. This is an optimization of the repeated verifier computation, not an algebraic shortcut: the initial product traverses the complete matrix through the same general routine also used for the in-domain unit-vector diagnostic.

The SHAKE256 adapter is not presented as an author encoding. Since the proof is pointwise for every target vector, no choice of random-oracle serialization can turn a Section 6.4-compliant threshold into a rejecting one. Likewise, the experiment's uniform public matrices are not trapdoor-certified keys, but $A0 = 0$ for every possible matrix. The reproducer therefore tests the claimed ordinary verifier relation without claiming key-generation conformance.

5.2 Independent full-row result

A fresh-context agent derived its implementation from the pinned paper before reading the primary experiment. It generated three independent full-size matrices and 64 messages per row. Table 2 summarizes the result.

Row	Messages	Canonical	Centered	Matrix SHA-256 prefix
TW-512-b2	64	64/64	64/64	`f447b22372e1b45e...`
TW-1024-b2	64	64/64	64/64	`aa0fa985706fef88...`
TW-1024-b4	64	64/64	64/64	`d9fa77d778166b3e...`

Table 2: Accepted zero-signature checks in the independent reproduction.

All 384 target checks accepted. The sequential run used one worker, 11.40 seconds wall time, 8.39 CPU seconds, and 61,072 KiB peak resident memory. It made zero signing queries and generated no honest signature. The independent results file has SHA-256 digest

`c350afc1310567ce31e93cd3b4456744fd8cabbe5ea17c34f105062d60533700.`

5.3 Precommitted diagnostics

The experiment does not use an invalid over-β vector or a threshold chosen from an observed sample. Its matrix-path diagnostic evaluates the well-formed in-domain vector e_0 and confirms that Ae_0 differs from $A0$. Distinct-message checks require every oracle vector and multiple residual norms to differ.

Figure 3's printed empirical $\|e\|_2$ values are preregistered as diagnostic radii only, never as α. In the independent 64-message run, the same well-formed zero signature produced the rejection counts in Table 3. These rejections show that the message hashing, lift, norm, and comparator paths are capable of rejecting in-domain inputs.

The polished public reproducer adds a deterministic target-space diagnostic. It uses the valid vector

$$u_{\mathrm{diag}} = (q/2, \ldots, q/2) \in \mathbb{Z}_q^n,$$

which must reject at the same printed diagnostic radius under both lifts. This vector and radius are fixed before any sample is drawn. It avoids a flaky or post-selected check while exercising the same residual-norm comparison.

Row	Canonical rejects	Centered rejects
TW-512-b2	64/64	32/64
TW-1024-b2	64/64	22/64
TW-1024-b4	64/64	59/64

Table 3: In-domain diagnostic rejections at the printed Figure 3 empirical $\|e\|_2$ radius. These values are not treated as verifier thresholds.

6 Specification inconsistencies and limitations

The following observations bound the result and should be resolved by any future specification:

1. **Missing thresholds.** Figure 3 publishes no concrete α, β, or σ. The result covers every completion obeying Section 6.4 but does not describe an unavailable alternative profile that ignores that section.
2. **Missing representation.** The paper specifies neither a modular lift nor hash, key, or signature serialization. Both natural fixed lifts pass; the proof holds for every hash vector.
3. **Private implementation.** The proof-of-concept is cited as private communication. No source replay or interoperability claim is made.
4. **Honest signing.** The audit did not attempt honest acceptance. Doing so would require inventing missing α, β, σ, χ, Ring mapping, and serialization details. The universal attack is independent of an honest signing distribution.
5. **Row-3 contradiction.** Figure 3 reports $s = 3989.3$ but $\|y\|_2 = 50533.7$ in the third "This work" row, contrary to a literal reading of Section 6.4's $\|x\|_2 \le s$ bound.
6. **Size mismatch.** Figure 2's full matrix public-key size formula does not reproduce Figure 3's kilobyte figures without an unstated structured representation.
7. **Empirical error norms.** The printed $\|e\|_2$ values are far below the Section 6.4 worst-case bound and are not identified as α. They cannot be silently substituted for the verifier threshold.

The result is a fresh-message forgery, not secret-key or trapdoor recovery. It says nothing about a future revision that changes the correctness analysis and threshold rule.

7 Prior work, attack history, and repair direction

Chen, Genise, and Mukherjee introduced Approx-ISIS and approximate trapdoors for smaller hash-and-sign signatures [2]. The zero-vector boundary is part of that generic Approx-ISIS context, and this report claims no novelty for it. The concrete finding is that the later higher-bit proposal's explicit Section 6.4 prescription crosses that boundary at each advertised row.

No prior public attack against the Higher-Bit Approximate-ISIS hash-and-sign signature of ePrint 2022/1144 was located in the searches performed as of 2026-10-14. This is a dated, bounded result and cannot exclude private, deleted, or unindexed work; it is not an absolute priority claim. The final search confidence is moderate-high.

A repair must make the verifier's accepted error region meaningfully smaller than the target space while retaining an honest-signature correctness margin. That requires revisiting the higher-bit approximation, signing-error analysis, and concrete thresholds together. Merely changing encodings, choosing a different lift, or rejecting the literal all-zero byte string has not been shown to restore unforgeability: a sound revision needs a complete specification and a new proof and full-parameter evaluation. This report does not validate a drop-in patch.

8 Reproducibility and provenance

The public artifact contains the complete standard-library source, three-row parameter transcription, deterministic quick and 64-message reference outputs, build files, this LaTeX source and PDF, and a SHA-256 manifest. Ordinary invocations draw a fresh 256-bit operating-system seed; reference invocations use disclosed deterministic fixtures. Every run is sequential with one worker and emits a machine-readable JSON record ending in the semantic banner `ATTACK REPRODUCTION: PASS` on success.

The technical path was checked in three roles: primary reproduction, hostile adversarial review,

and fresh-context independent reproduction. The final exact-target history recheck and independent Publication Review also passed. The package retains the stricter claim boundary required by those reviews and is classified as `VERIFIED_ATTACK`. No independent human verification was performed or required.

The final-history handoff has SHA-256 digest

`daadbc0743cb5ff1c76d2b1cd842261157e3dd438f06bc35579ee1529431c81b,`

and its sealed evidence manifest has SHA-256 digest

`9702681ceaaeb90882e0d04c9058f4558fe3f0a5412be2211bd3ede6212d4657.`

Acknowledgements

The computations were performed on the Norwegian Research and Education Cloud (NREC), using resources provided by the University of Bergen and the University of Oslo.

References

[1] Anaëlle Le Dévéhat, Hiroki Shizuya, and Shingo Hasegawa. On the Higher-bit Version of Approximate Inhomogeneous Short Integer Solution Problem. IACR Cryptology ePrint Archive, Report 2022/1144, 2022. https://eprint.iacr.org/2022/1144.

[2] Yilei Chen, Nicholas Genise, and Pratyay Mukherjee. Approximate Trapdoors for Lattices and Smaller Hash-and-Sign Signatures. In *Advances in Cryptology–ASIACRYPT 2019*, LNCS 11923, pages 3–32. Springer, 2019. https://doi.org/10.1007/978-3-030-34618-8_1.